BAZAARLENS PK

Pakistan Grocery Price Intelligence & City-Level Analysis

FINAL PROJECT REPORT

A Python-based analysis of grocery prices, city-level variation, weekly price movements, and a fixed household grocery basket.

Project: BazaarLens PK

Prepared: November 2026

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter Notebook

Prepared as a data analytics portfolio project

Executive Summary

BazaarLens PK is a data analytics portfolio project that explores grocery prices in Pakistan using one Kaggle dataset and two Pakistan Bureau of Statistics (PBS) datasets. The project examines item-level prices, city-wise averages, short-term weekly movements, and the cost of a fixed basket of 11 selected grocery items.

The analysis combines 350 Kaggle records collected on 28 April 2025 across five cities, a PBS monthly dataset covering 51 essential items across 17 cities for September 2026, and five PBS weekly reports covering October 2026. The work was organized into eight Jupyter notebooks, from data understanding and cleaning through analysis, visualization, and final insights.

Headline findings

• Islamabad recorded the highest city average in all three datasets, although the datasets differ in coverage, time period, and structure.

• In the Kaggle dataset, the highest average item price was Eggs (Dozen) at Rs. 396.02, while Moong Dal had the lowest at Rs. 243.84. Units differ, so these values are not standardized comparisons.

• In the PBS weekly comparison of four selected items, Chicken Farm Broiler (Live) increased by 18.02% and Wheat Flour Bag (20 Kg) increased by 9.42% across the five weekly observations.

• The fixed 11-item basket cost was highest in Islamabad at Rs. 8,226.68 and lowest in Faisalabad at Rs. 7,164.49, a difference of Rs. 1,062.19.

The results are descriptive. They identify observed differences and changes but do not establish causes, represent every household's spending, or constitute an official inflation index.

Report Contents

Section	Description
1	Project overview, objectives and scope
2	Data sources and dataset descriptions
3	Tools and analytical methodology
4	Data understanding and cleaning
5	Exploratory data analysis and Kaggle findings
6	PBS monthly and city-wise findings
7	PBS weekly and selected-item trends
8	Fixed grocery basket analysis
9	Visualizations and key findings
10	Recommendations, limitations and conclusion

1. Project Overview

BazaarLens PK focuses on understanding how recorded grocery prices vary by item, city, and reporting period in Pakistan. It brings together data from different sources to demonstrate a practical end-to-end data analytics workflow using Python.

1.1 Objectives

• Inspect and understand grocery price datasets before analysis.

• Clean and prepare Kaggle and PBS data for reliable analysis.

• Explore item-wise price distributions and identify notable values.

• Compare average recorded prices across Pakistani cities.

• Examine weekly price movements for selected essential commodities.

• Calculate and compare the cost of a fixed basket of 11 selected items.

• Communicate findings through charts, summaries, and recommendations.

1.2 Scope

The project uses a single-date Kaggle snapshot from April 2025, one PBS monthly snapshot for September 2026, and five PBS weekly reports from October 2026. It is a comparative descriptive analysis rather than a continuous national price-monitoring system.

1.3 Questions explored

• Which grocery items had the highest and lowest average recorded prices in the Kaggle dataset?

• How did average recorded prices differ among cities within each dataset?

• Which selected items showed the largest weekly price increases or decreases?

• How did the cost of the selected 11-item basket vary among the 17 cities?

2. Data Sources and Dataset Description

The project uses three datasets. Their reporting periods, item coverage, city coverage, and structure are different, so results are interpreted within each dataset rather than treated as directly interchangeable.

Dataset	Coverage	Reporting period / structure
Kaggle – Pakistan City Wise Grocery Prices	350 records; 15 items; 5 cities	Prices collected 28 April 2025
PBS Monthly	51 items; 17 cities; 25 columns	September 2026 monthly snapshot, with comparison fields
PBS Weekly	Five weekly reports; 51 items per report; 17 cities	1, 8, 15, 22 and 29 October 2026

2.1 Kaggle dataset fields

The Kaggle dataset includes City, Item, Brand, Price_per_Kg_Litre, Date_Collected, and Market_Source. The data covers Karachi, Quetta, Islamabad, Lahore, and Peshawar, and includes sources such as Local Bazaar, Imtiaz, Metro, Al-Fatah, and Chase Up.

2.2 PBS monthly dataset

The PBS monthly file contains 51 item rows and 17 city price columns, along with national/monthly/yearly average and change fields. It includes September 2026 prices and comparison fields for earlier periods. Three missing values were observed for Rice IRRI-6/9 in Gujranwala, Sialkot, and Lahore.

2.3 PBS weekly dataset

The weekly dataset combines five weekly Excel reports into 255 rows and 66 columns. It contains city-level minimum, average, and maximum price fields and national average fields. Missing values for Toilet Soap LIFEBUOY 115 gm in the first three weekly dates across 20 columns were retained rather than filled with invented values.

3. Tools and Analytical Methodology

The analysis was carried out in Python using Jupyter notebooks. The project uses the following tools and libraries:

• Python – programming and analysis environment

• Pandas – loading, inspecting, cleaning, grouping, and summarizing tabular data

• NumPy – numerical operations

• Matplotlib and Seaborn – statistical and categorical visualizations

• Plotly Express – interactive line and comparison charts

• Jupyter Notebook – documenting the analysis workflow

3.1 Analytical workflow

Stage	Work performed
Data understanding	Loaded files, reviewed dimensions, columns, data types, and summary statistics.
Data cleaning	Standardized column names and text values, converted dates and numeric fields, checked missing values, duplicates, a
Exploratory analysis	Reviewed distributions, summary statistics, item-level averages, and potential outliers.
City-wise analysis	Calculated city-level averages separately for Kaggle, PBS Monthly, and PBS Weekly.
Time-series analysis	Compared five weekly observations for selected PBS items.
Basket analysis	Summed the prices of 11 selected grocery items for each of 17 cities.
Visualization and insights	Created charts and summarized findings, limitations, and recommendations.

3.2 Data quality approach

Missing values were inspected and retained where a reliable replacement could not be justified. Duplicate records and zero or negative Kaggle prices were checked. Dates were converted to datetime formats, and city/item text was standardized where required. No missing values or duplicate records were found in the cleaned Kaggle file; no zero or negative Kaggle prices were identified.

3.3 Outlier method

The Kaggle price distribution was assessed using the interquartile range (IQR). Q1 was Rs. 198.695, Q3 was Rs. 451.6625, and IQR was Rs. 252.9675. The calculated upper boundary was Rs. 831.11375 and lower boundary was Rs. -180.75625. No records exceeded these IQR boundaries. This does not prove that every value is error-free; it only means the rule did not flag any observations.

4. Project Notebook Structure

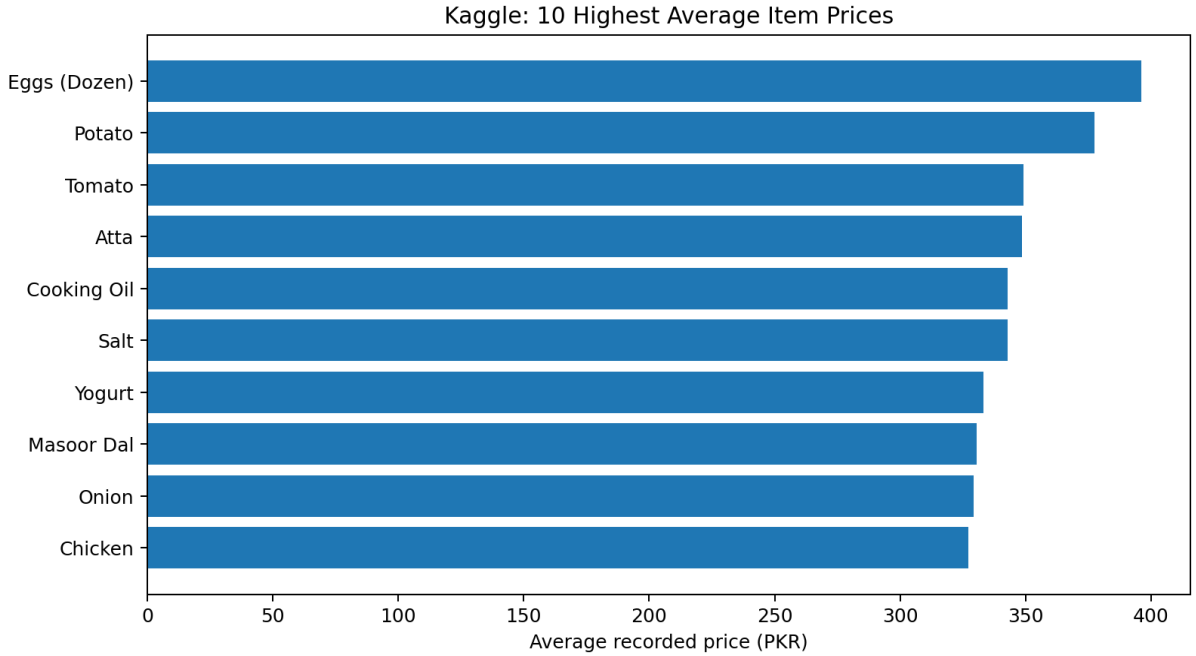
Notebook	Purpose
01_Data_Understanding.ipynb	Initial inspection and understanding of the raw datasets.
02_Data_Cleaning.ipynb	Cleaning, validation, type conversion, and saving cleaned datasets.
03_Exploratory_Data_Analysis.ipynb	Distributions, descriptive statistics, item-level analysis, and IQR checks.
04_City_Wise_Analysis.ipynb	City-level average comparisons across the three datasets.
05_Time_Series_Analysis.ipynb	Weekly price movement analysis for selected items.
06_Grocery_Basket_Analysis.ipynb	Cost of the fixed 11-item grocery basket across 17 cities.
07_Data_Visualization.ipynb	Consolidated visual presentation of important findings.
08_Final_Insights.ipynb	Final insights, recommendations, conclusion, and limitations.

5. Exploratory Data Analysis – Kaggle

The Kaggle dataset contains 350 records across 15 grocery items and five cities. All records share the collection date 28 April 2025, so the dataset provides a snapshot rather than a time series.

5.1 Item-wise average recorded prices

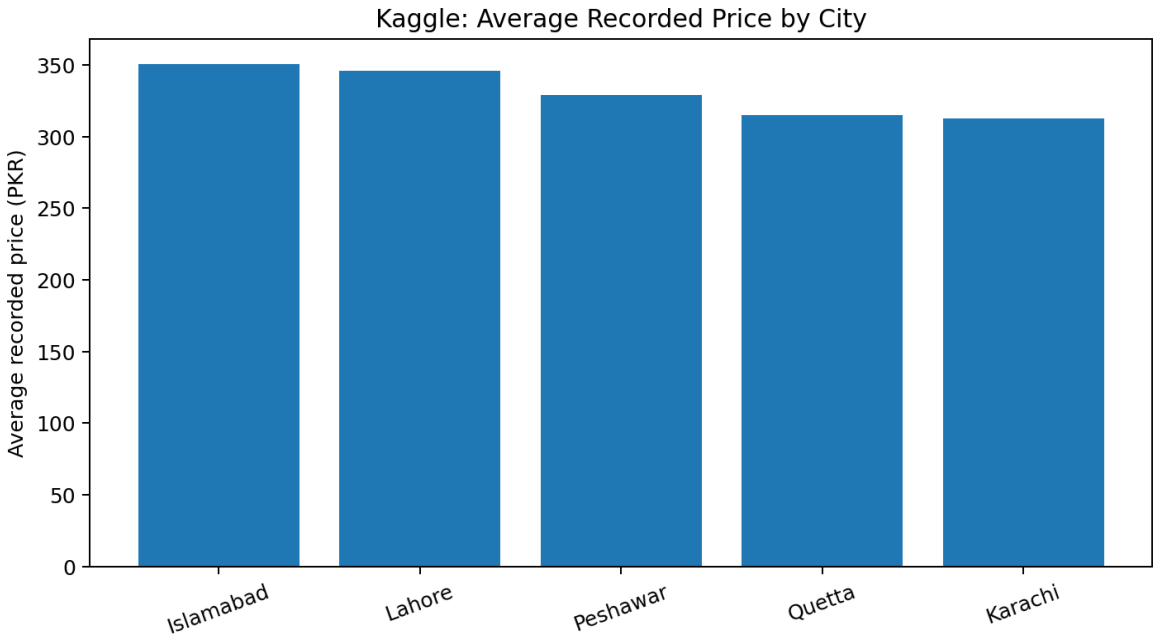
Item	Average recorded price (PKR)
Eggs (Dozen)	396.01
Potato	377.58
Tomato	349.20
Atta	348.62
Cooking Oil	342.98
Salt	342.70
Yogurt	333.23
Masoor Dal	330.51
Onion	329.34
Chicken	327.34
Rice	315.98
Tea	315.96
Sugar	314.75
Milk	284.97
Moong Dal	243.84



Eggs (Dozen) had the highest average recorded price at Rs. 396.02, followed by Potato at Rs. 377.58. Moong Dal had the lowest average recorded price at Rs. 243.84. These are observed averages from the dataset.

Unit caution: Eggs are priced per dozen while most other items are recorded per kilogram or litre. Therefore, ranking these values does not compare equivalent quantities.

5.2 Kaggle city-wise analysis



City	Average recorded price (PKR)
Islamabad	350.33
Lahore	345.65
Peshawar	329.24
Quetta	315.27
Karachi	312.41

Islamabad recorded the highest average at Rs. 350.33, while Karachi recorded the lowest at Rs. 312.41. The difference was approximately Rs. 37.92.

5.3 Kaggle data quality findings

• 350 records and 15 unique grocery items were included.

• No missing values or duplicate records were found after cleaning.

• No zero or negative grocery prices were identified.

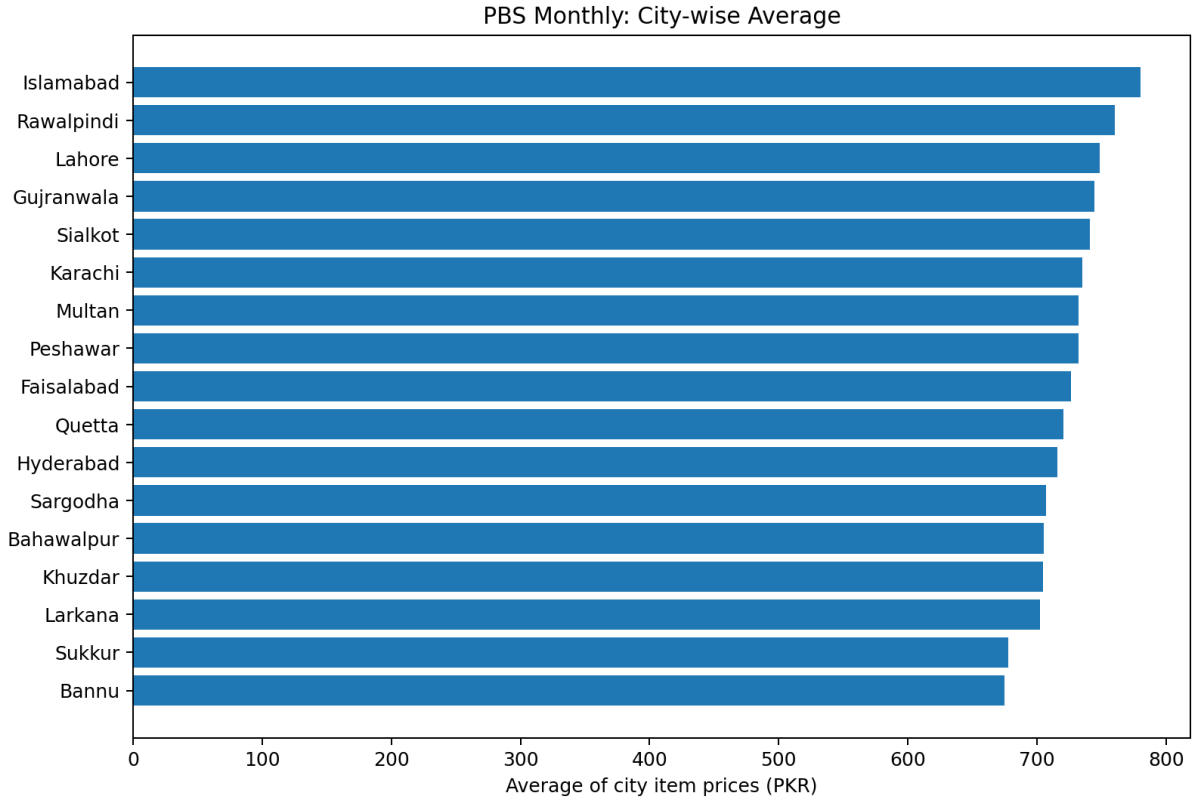
• The IQR check did not flag any price outliers.

Interpretation: The Kaggle data describes prices collected on one date from five cities and should not be used to infer inflation or historical price movements.

6. PBS Monthly Analysis

The PBS monthly dataset provides a September 2026 snapshot of 51 essential items across 17 cities. City averages are calculated from the available item-level city prices and are descriptive summaries, not a household cost index.

6.1 City-wise average item prices



City	Average item price (PKR)
Islamabad	779.92
Rawalpindi	760.05
Lahore	748.26
Gujranwala	744.14
Sialkot	740.80
Karachi	735.26
Multan	732.27
Peshawar	732.12
Faisalabad	726.39
Quetta	720.66
Hyderabad	715.55
Sargodha	707.22

City	Average item price (PKR)
Bahawalpur	704.93
Khuzdar	704.60
Larkana	702.33
Sukkur	677.66
Bannu	674.84

Islamabad recorded the highest city average at Rs. 779.92. Bannu recorded the lowest at Rs. 674.84. The difference was approximately Rs. 105.08.

6.2 Item and price-change observations

The monthly visualization identified high-priced items including the LPG 11.67 kg cylinder, a 5-litre tin of DALDA or similar cooking oil, and a wheat flour bag. Separate charts examined the largest monthly and yearly percentage price increases. Exact item-by-item change values were not available in the project summary used to prepare this report, so no unverified percentages are reproduced here.

6.3 Data quality notes

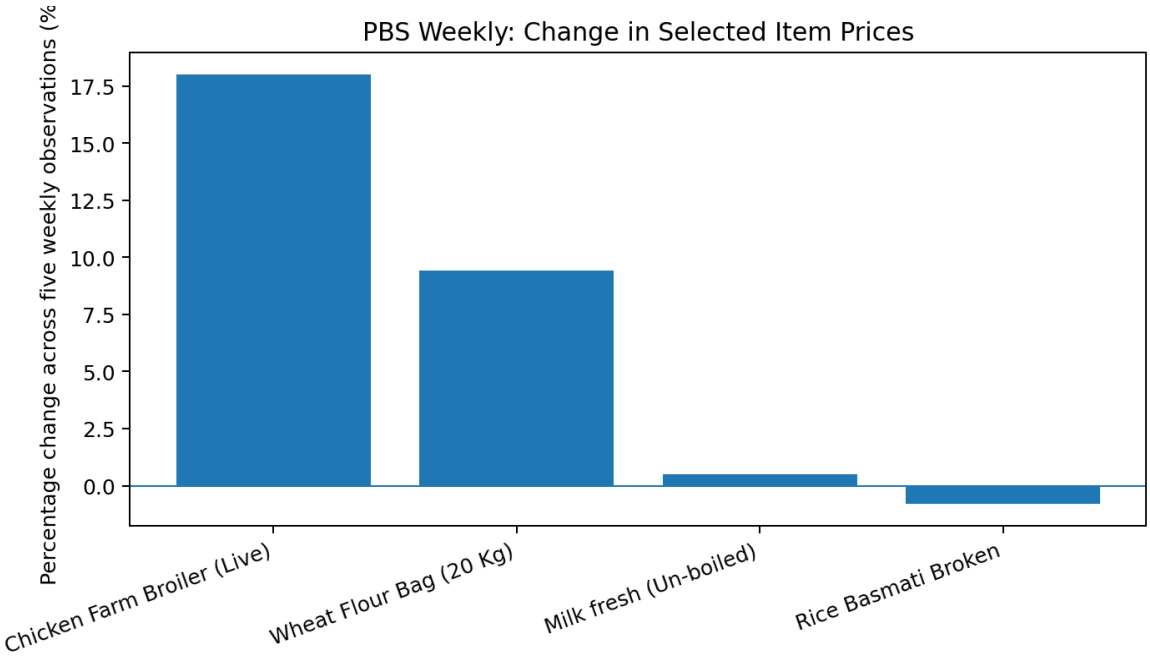
Three missing values were retained for Rice IRRI-6/9 in Gujranwala, Sialkot, and Lahore. The monthly file also contains comparison fields for monthly and yearly changes. It represents one monthly reporting period in the supplied analysis and does not provide a long historical monthly series.

7. PBS Weekly Analysis

The weekly dataset combines five reports dated 1, 8, 15, 22, and 29 October 2026. It contains 51 essential items per weekly report and city-level price fields for 17 cities.

7.1 Selected item price movements

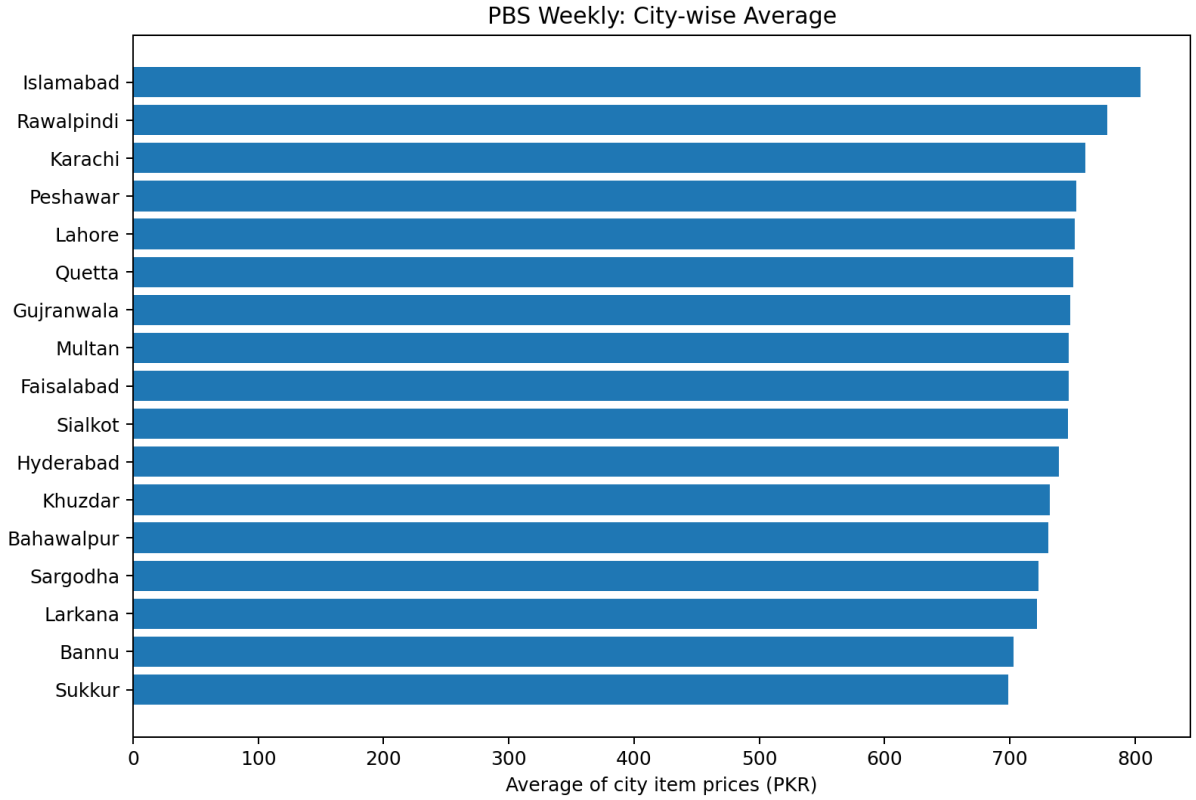
Selected item	Change across five weekly observation
Chicken Farm Broiler (Live)	+18.02%
Wheat Flour Bag (20 Kg)	+9.42%
Milk fresh (Un-boiled)	+0.50%
Rice Basmati Broken	-0.80%



Chicken Farm Broiler (Live) showed the largest increase among the four selected items at 18.02%. Wheat Flour Bag (20 Kg) increased by 9.42%. Milk fresh (Un-boiled) increased by 0.50%, while Rice Basmati Broken (Average Quality) decreased by 0.80%.

For two selected items, the first and last weekly national average values were available in the project notes: wheat flour moved from Rs. 2,697.45 to Rs. 2,951.46, and live broiler chicken moved from Rs. 365.31 to Rs. 431.15. Both reached a higher value on 22 October before a small decline on 29 October.

7.2 City-wise weekly average


PBS Weekly: City-wise Average
Islamabad
Rawalpindi
Karachi
Peshawar
Lahore
Quetta
Gujranwala
Multan
Faisalabad
Sialkot
Hyderabad
Khuzdar
Bahawalpur
Sargodha
Larkana
Bannu
Sukkur
0
100
200
300
400
500
600
700
800
Average of city item prices (PKR)

7.2 City-wise weekly average – values

City	Average item price (PKR)
Islamabad	804.21
Rawalpindi	777.89
Karachi	760.18
Peshawar	752.86
Lahore	751.97
Quetta	750.41
Gujranwala	748.23
Multan	747.15
Faisalabad	746.79
Sialkot	746.42
Hyderabad	739.33
Khuzdar	731.83
Bahawalpur	730.72
Sargodha	722.83
Larkana	721.62
Bannu	702.83
Sukkur	698.99

Islamabad recorded the highest average at Rs. 804.21, while Sukkur recorded the lowest at Rs. 698.99. The difference was approximately Rs. 105.22.

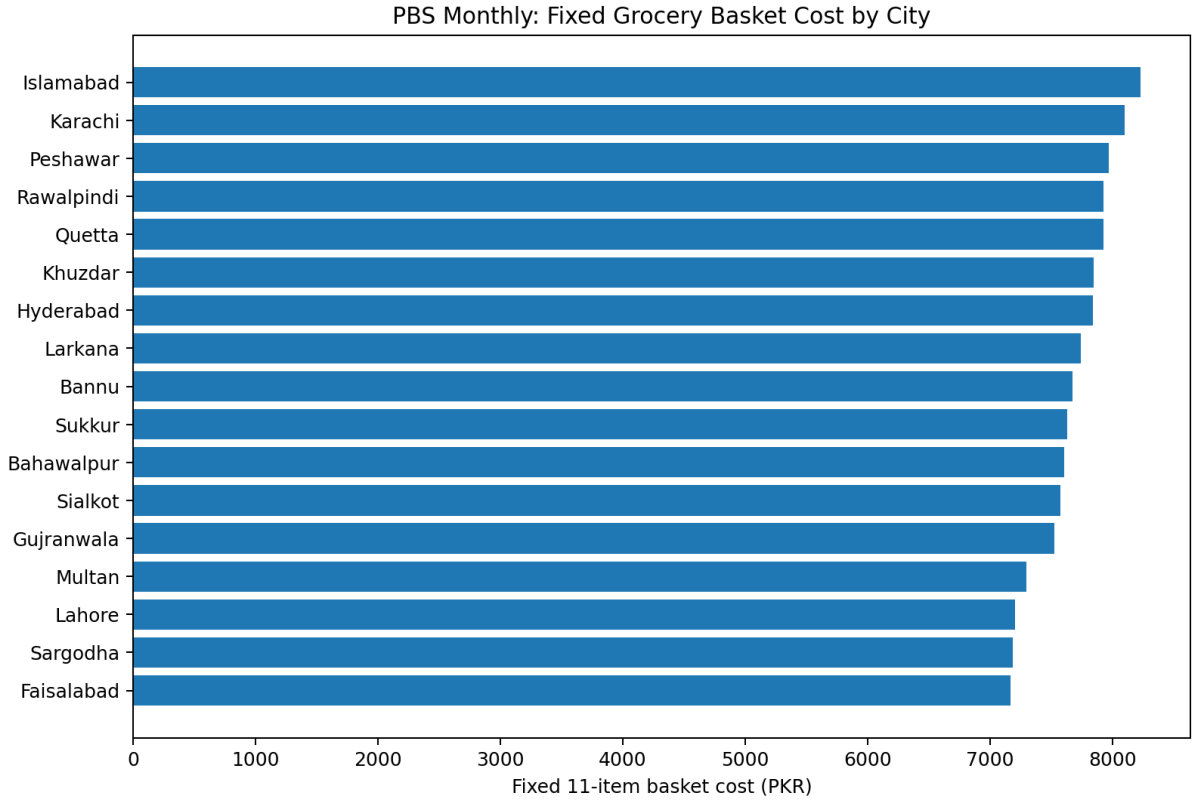
7.3 Weekly data notes

The five-week window is short and does not support conclusions about long-term inflation or seasonal behavior. Weekly city averages summarize item-level values and should not be interpreted as a standardized cost-of-living measure. Missing observations in the source reports were retained rather than fabricated.

8. Fixed Grocery Basket Analysis

The project calculated the combined price of a fixed basket of 11 selected essential grocery items for each of 17 cities using PBS monthly data. The basket includes wheat flour, rice, chicken, milk, eggs, cooking oil, masoor pulse, potatoes, onions, tomatoes, and sugar.

8.1 City-wise basket cost



City	Basket cost (PKR)
Islamabad	8,226.68
Karachi	8,098.75
Peshawar	7,971.08
Rawalpindi	7,928.89
Quetta	7,924.62
Khuzdar	7,848.29
Hyderabad	7,840.33
Larkana	7,739.16
Bannu	7,672.62
Sukkur	7,628.60
Bahawalpur	7,604.31
Sialkot	7,576.94

City	Basket cost (PKR)
Gujranwala	7,524.54
Multan	7,295.39
Lahore	7,202.21
Sargodha	7,183.53
Faisalabad	7,164.49

Islamabad had the highest selected basket cost at Rs. 8,226.68. Faisalabad had the lowest at Rs. 7,164.49. The difference was Rs. 1,062.19.

8.2 Five highest and five lowest basket costs

Highest-cost cities	Cost (PKR)	Lowest-cost cities	Cost (PKR)
Islamabad	8,226.68	Faisalabad	7,164.49
Karachi	8,098.75	Sargodha	7,183.53
Peshawar	7,971.08	Lahore	7,202.21
Rawalpindi	7,928.89	Multan	7,295.39
Quetta	7,924.62	Gujranwala	7,524.54

The basket comparison shows variation in the cost of this selected group of items across cities. It is not a complete household expenditure basket, and the result depends on the 11 chosen items and their PBS units.

9. Consolidated Key Findings

Area	Finding
Kaggle item analysis	Eggs (Dozen) had the highest average recorded price (Rs. 396.02); Moong Dal had the lowest (Rs. 243.84).
Kaggle city analysis	Islamabad highest (Rs. 350.33); Karachi lowest (Rs. 312.41).
PBS Monthly city analysis	Islamabad highest (Rs. 779.92); Bannu lowest (Rs. 674.84).
PBS Weekly city analysis	Islamabad highest (Rs. 804.21); Sukkur lowest (Rs. 698.99).
Weekly selected items	Chicken +18.02%; wheat flour +9.42%; milk +0.50%; broken basmati rice -0.80%.
Fixed grocery basket	Islamabad highest (Rs. 8,226.68); Faisalabad lowest (Rs. 7,164.49); gap Rs. 1,062.19.

10. Recommendations

10.1 For consumers

• Compare prices among local markets before purchasing where practical.

• Pay attention to commodities with notable observed weekly increases, such as chicken and wheat flour.

• Use a consistent household shopping list to monitor changes in personal grocery spending.

10.2 For retailers

• Monitor supplier and market prices for essential commodities on a regular schedule.

• Use item-level price tracking to support purchasing and inventory planning.

• Review local market conditions before making pricing decisions.

10.3 For policymakers and data providers

• Continue publishing consistent, timely city-wise price information.

• Improve completeness and consistency of reported item units and city coverage.

• Monitor commodities showing substantial short-term price movements and communicate reliable price information to consumers.

These are practical recommendations suggested by the descriptive findings. The analysis does not establish the causes of price changes or evaluate the effectiveness of any policy.

11. Limitations and Future Improvements

11.1 Limitations

• The Kaggle dataset contains one collection date (28 April 2025), so it cannot show historical trends.

• The PBS monthly dataset used in this project represents September 2026 rather than a long monthly history.

• The PBS weekly analysis covers only five weeks in October 2026.

• Weekly trend comparison focuses on four selected items, not all items in the dataset.

• The datasets differ in coverage, item definitions, units, and reporting periods; direct cross-dataset numerical comparisons require caution.

• Some grocery items use different units, such as per dozen, per kilogram, per litre, or per pack. Their prices are not directly comparable without unit normalization.

• The fixed basket includes 11 selected items and does not represent every household's full grocery expenditure.

• Some PBS missing values were retained, which may affect averages where the relevant item/city observation is absent.

• The project is descriptive and does not identify causal drivers of price changes or calculate an official inflation index.

11.2 Future improvements

• Add longer historical PBS monthly and weekly datasets to study trends over time.

• Expand the analysis to more items, markets, and locations where reliable data is available.

• Normalize units to support more meaningful like-for-like item comparisons.

• Create an interactive dashboard using Power BI or Tableau.

• Explore forecasting methods after collecting a sufficiently long and consistent time series.

• Automate data updates and quality checks to support repeatable analysis.

12. Conclusion

BazaarLens PK demonstrates an end-to-end Python data analytics workflow applied to grocery price information in Pakistan. Across the three datasets, the project inspected and cleaned data, explored item-level distributions, compared city-level averages, reviewed short-term weekly movements, calculated a fixed grocery basket, and communicated results through visualizations and written insights.

The findings show observable differences in recorded prices across items and cities. Islamabad had the highest city average in each of the three dataset-specific comparisons, while the lowest city differed by dataset. In the five-week PBS analysis, chicken and wheat flour showed the largest increases among the four selected items. The fixed 11-item basket was most expensive in Islamabad and least expensive in Faisalabad, with a difference of Rs. 1,062.19.

The project provides a foundation for further price monitoring and dashboard development. Its conclusions should be understood within the available data periods, units, and coverage, and should not be interpreted as a complete measure of grocery affordability or national inflation.

End of Report

BazaarLens PK | Final Project Report | November 2026